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Virtual reality as an engagement driver in professional meetings

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Veronika Šlapáková Losová and Marko Orel
Department of Entrepreneurship, Prague University of Economics and Business, Prague, Czech Republic

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Abstract

Purpose – This study proposes how to empirically measure and explain virtual reality's (VR) impact on meeting engagement compared to traditional video conferencing through a novel methodological framework grounded in social presence theory, combining social network analysis with conversation analysis to address the lack of empirical evidence on VR's effectiveness in authentic workplace settings.

Design/methodology/approach – Through a 16-week longitudinal experiment in a multinational company, we analyzed eight retrospective meetings to compare engagement patterns across platforms. The framework operationalizes engagement through group cohesion metrics and content characteristics, employing both quantitative network analysis and qualitative content analysis of interaction patterns.

Findings – Results suggest that VR meetings generate higher engagement levels through three key mechanisms: enhanced network density, more balanced participation patterns, and improved social presence behaviors. The most striking finding reveals VR's capacity to enhance peripheral participant engagement, demonstrated by an increase in degree prestige for least active participants. Additionally, while VR meetings exhibited lower task-orientation, they fostered more dynamic and inclusive interactions, challenging traditional assumptions about meeting efficiency as a function of task-orientation.

Originality/value – The study presents empirical evidence of VR's impact on meeting engagement in authentic workplace settings through an innovative methodological framework. The integration of social network analysis with conversation analysis establishes a non-intrusive method for evaluating meeting effectiveness, capturing both structural and qualitative aspects of engagement. These findings advance our understanding of how VR technology transforms interaction dynamics, suggesting a paradigm shift in virtual collaboration practices.

Keywords Virtual reality, Meeting engagement, Social network analysis, Social presence theory, Virtual collaboration

Paper type Research article

1. Introduction

> *Virtual reality, virtual space, virtual organizations, virtual teams; the word "virtual" is today's organizational buzzword. Virtual is there but not there, and this is exactly what Alice experienced with the Cheshire Cat in Wonderland. So, the virtual world really is not as new an idea as some of us might have thought. One of the fastest-growing, high-tech office trends today is "virtual teams".*
>
> - Johnson *et al.* (2001), p. 24.

More than two decades after Johnson and colleagues highlighted the emergence of virtual teams, organizations still grapple with understanding and measuring workers' engagement in virtual environments. While virtual reality (VR) technology has evolved substantially, fundamental questions about its impact on meeting dynamics and participant engagement remain largely unanswered. Research has established that participant engagement significantly influences the effectiveness of virtual meetings (Frank *et al.*, 2016; Leach *et al.*, 2009), with Leach *et al.* (2009) specifically identifying participant involvement as a determining mediator between meeting characteristics and perceived effectiveness.

Traditional methods of measuring this engagement have typically followed two approaches: self-reported surveys (Przybylek and Kotecka, 2017; Leach *et al.*, 2009) and laboratory-based measurements of non-verbal indicators using sensors (Frank *et al.*, 2016). While surveys are limited by their reliance on subjective interpretations and individual feelings, sensor-based measurements, though providing detailed and objective data, are confined to staged experiments that may affect results and remain inaccessible to companies and researchers. Both methods demand additional time and effort from participants, making

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them impractical for real-world implementation. As Anderson *et al.* (2007) pointedly observe, this limitation is evident in how most virtual collaboration studies are conducted in laboratory settings, often using ad hoc student teams working on artificial tasks that poorly reflect real-world scenarios.

The rather swift adoption of immersive computer technologies in organizational contexts represents a significant shift in how businesses approach collaboration and communication (Masood and Egger, 2019; Orel, 2022). While traditional video-conferencing platforms have served as the primary medium for remote interaction, they often fail to create genuine engagement and presence among participants (Bailenson, 2021). VR-supported/based meetings, by contrast, offer an immersive three-dimensional environment that potentially overcomes these limitations by simulating physical presence and enabling natural interactions through avatars (Di Pasquale *et al.*, 2022).

Despite these promising attributes, empirical research examining the effectiveness of VR meetings in organizational settings remains surprisingly limited (Orel, 2022). Most existing studies have relied on controlled laboratory experiments with student participants or self-reported surveys, leaving a significant gap in our understanding of how VR meetings function in authentic business environments (Frank *et al.*, 2016; Leach *et al.*, 2009). This gap becomes particularly critical to justify investments and guide implementation as organizations increasingly invest in VR technologies (Kaplan *et al.*, 2021).

The following study examines how VR technology affects meeting engagement compared to traditional video conferencing, using a methodological framework grounded in social presence theory (SPT). Through a longitudinal experiment conducted with a multinational company, we analyze eight retrospective meetings to demonstrate that VR meetings generate significantly higher levels of engagement than video conferences. Our contribution is threefold: First (1), we provide empirical evidence of VR's capacity to enhance meeting engagement through three core mechanisms—enhanced group cohesion, strengthened social presence behaviors, and dynamic interaction balance—while demonstrating how embodied interaction promotes equity of voice. Second (2), we demonstrate how the combination of social network analysis and conversation analysis can reveal nuanced patterns of engagement in virtual meetings, capturing both structural and qualitative aspects of participant interaction. Third (3), we offer practical insights for organizations considering VR technology adoption, showing how different meeting modalities influence participant engagement patterns and team dynamics. With that in mind, this research addresses a critical gap in understanding how (still) emerging (immersive) technologies such as VR impact workplace interaction, moving beyond laboratory (controlled) studies to examine real-world implementation in professional settings.

2. Literature review

2.1 Virtual meeting and engagement challenges

The landscape of organizational meetings has undergone a sizeable transformation in recent years, revealing both persistent challenges and emerging complexities. Research indicates that managers dedicate up to 80% of their workday to meetings, with half of this time potentially being unproductive (Mroz *et al.*, 2018). The (great) disruption behind the COVID-19 pandemic accelerated this trend, triggering a 13% increase in meeting frequency while reducing average duration by 20% (DeFilippis *et al.*, 2020). This compression of meeting time and the rapid shift to virtual formats have created unprecedented pressure on organizations to maximize meeting effectiveness. While meetings remain fundamental to organizational success—serving as crucial forums for information exchange, team cohesion, innovation, and problem-solving—the virtual transformation has introduced new barriers to these essential functions. Recent studies indicate that the challenges extend beyond mere technological adaptation, encompassing fundamental changes in how teams interact, collaborate, and maintain engagement in digital environments (Standaert *et al.*, 2022).

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The transition to predominantly virtual meetings has generated a cascade of interconnected challenges that threaten organizational effectiveness. Video conferences, while providing a functional platform for remote interaction, have been associated with increased cognitive load, heightened stress levels (Kumar *et al.*, 2023; Standaert *et al.*, 2022; Luong and Rogelberg, 2005), and a distinct form of fatigue that impacts both individual and team performance (Epstein, 2020; Hacker *et al.*, 2020). This phenomenon extends beyond simple screen fatigue, manifesting in reduced participation, diminished creative thinking, and weakened team dynamics. Particularly concerning is the emergence of "virtual presenteeism"—where participants are physically present but mentally disengaged, often multitasking or experiencing decreased motivation due to the perceived artificiality of digital interactions (Ruhle and Schmoll, 2021; Nurmi and Pakarinen, 2023). The latter deterioration in meeting effectiveness poses a significant challenge for organizations, particularly as they seek to maintain team cohesion and productivity in increasingly virtual work environments.

2.2 Meeting engagement and its evaluation

Meeting engagement has emerged as a critical determinant of meeting effectiveness across multiple contexts. Originally conceptualized in broader workplace settings, where Schaufeli and Salanova (2006) demonstrated its influence on project outcomes, engagement has been increasingly recognized as fundamental to meeting success. Recent research establishes that engagement directly impacts meeting quality, productivity, and overall effectiveness (Frank *et al.*, 2016; Panteli *et al.*, 2019), while extending beyond immediate meeting outcomes to influence broader organizational behaviors (Huang *et al.*, 2022). In virtual and dispersed team contexts, engagement becomes particularly decisive for effective functioning, interacting with factors such as attendance patterns, parallel work management, and awareness of others' presence (Mark *et al.*, 1999) – or what Davies (2004) would call "presence of the present".

Generally speaking, the literature distinguishes between two fundamental modes of engagement in meetings: *intrinsic* and *extrinsic*. Intrinsic engagement manifests through attentive listening, active observation, and responsive participation, demonstrating intention to contribute meaningfully (Frank *et al.*, 2016). It encompasses vigor, dedication, absorption in meeting activities (Schaufeli and Salanova, 2006), and participants' perceived value of collective interaction (Salam and Chetouani, 2015). Extrinsic engagement is characterized by observable behaviors such as connections with other participants (Williams *et al.*, 2019), demonstrated through mutual interaction patterns, meaningful involvement, and active participation in decision-making processes (Yoerger *et al.*, 2015). This "dual-mode" conceptualization has been further expanded by scholars such as Huang *et al.* (2022), who propose an overarching framework encompassing emotional, cognitive, and physical dimensions of engagement.

Traditional approaches to measuring meeting engagement broadly fall into two categories: *participant-focused* and *sensor-based* approaches. The participant-focused approach relies on self-reported, post-hoc measures (Vedernikov *et al.*, 2024), such as self-declaratory surveys measuring communication improvements and involvement (Przybylek and Kotecka, 2017; Leach *et al.*, 2009). However, these methods often suffer from social desirability bias (Fisher, 1993) and survey fatigue (Porter *et al.*, 2004; Galesic and Bosnjak, 2009), while typically capturing only the most engaged participants' experiences. The sensor-based approach employs technologies such as ECG and computer-vision techniques to measure participants' biological states (Frank *et al.*, 2016; Vedernikov *et al.*, 2024) but faces implementation challenges in distributed team settings and requires specialized equipment.

These constraints have led to a predominance of studies conducted with ad hoc student teams in laboratory settings, often addressing artificial tasks (Anderson *et al.*, 2007), while research on real-world meetings with established teams remains scarce (Olson *et al.*, 1992). Furthermore, both survey and sensor-based approaches focus primarily on individual engagement, overlooking team-level dynamics. This gap is significant, mainly as group

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dynamics are vital for virtual team performance (Driskell *et al.*, 2003), and interactive small-group processes contribute to team cohesion and entitativity (Blanchard and McBride, 2020). While some scholars have employed content analysis (Olson *et al.*, 1992; Bales, 1950) and quantitative interaction analysis (Anderson *et al.*, 2007) to study group dynamics, these methods have not been extensively applied to analyze engagement in professional meetings. To address these limitations and capture the complex nature of engagement in virtual meetings, researchers increasingly recognize the need for theoretical frameworks that can accommodate both individual and group-level dynamics while remaining practical for real-world implementation, with the SPT offering a particularly "promising" foundation for understanding and measuring engagement in virtual (work) settings.

2.3 Social presence theory

SPT, originally developed by Short *et al.* (1976), has evolved markedly from its initial focus on communication media properties that induce feelings of intimacy (Argyle and Dean, 1965) and immediacy (Wiener and Mehrabian, 1968). While initially conceived to explain differences in presence across various communication media, the theory has expanded to address subjective experiences of co-presence in diverse contexts, including meetings, virtual environments, and educational settings (Biocca *et al.*, 2003). Contemporary understanding defines social presence "simply" as the sense of "being with another" or the degree to which a person is perceived as "real" in mediated communications (Gunawardena and Zittle, 1997).

In technology-mediated environments, social presence manifests across three distinct levels. At the perceptual level, spatial co-presence focuses on the basic awareness of others in the shared virtual space. The subjective level concerns psychological perceptions of presence and the assessed accessibility of others, moving beyond physical presence to encompass intellectual co-presence that enables psychological involvement and behavioral engagement (Biocca *et al.*, 2003). The highest level, intersubjective presence, emerges through dynamic interactions between participants, where behavior becomes interdependent, connected, and responsive to others (Cui *et al.*, 2012). This behavioral engagement component is considered the pinnacle of social presence, building upon the foundation of co-presence and psychological involvement.

The relationship between social presence and meeting engagement is both reciprocal and reinforcing. Kreijns *et al.* (2022) establish that social interactions are both a function of social presence and a mechanism for strengthening it. This dynamic is particularly relevant in virtual meetings, where social presence is not binary but exists on a continuum of perceived connection and involvement (Biocca *et al.*, 2003). Behavioral indicators of engagement, such as responsiveness and active participation, serve as visible manifestations of social presence (Sivunen and Nordback, 2014), making them valuable metrics for assessing meeting effectiveness. Social presence has proven crucial for facilitating a positive social climate (Cui *et al.*, 2012), enhancing group dynamics, and fostering social cohesion (Kreijns *et al.*, 2022)—elements that are essential for successful virtual meetings. The research consistently shows that social presence positively impacts engagement in the workplace context through social support, organizational identification and interaction with leaders and colleagues (Ha *et al.*, 2023; Simbula *et al.*, 2023).

2.4 VR in professional settings

The accelerating adoption of immersive technologies in organizational contexts marks a significant evolution in workplace collaboration (Masood and Egger, 2019; Orel, 2022). While traditional video-conferencing has become the default medium for remote interaction, research increasingly highlights its limitations in fostering genuine participant engagement and presence (Bailenson, 2021). VR addresses these limitations by creating immersive three-dimensional environments that simulate physical presence and enable natural interactions through avatars (Di Pasquale *et al.*, 2022). Recent advances in embodiment research demonstrate that avatar-mediated interaction in VR environments enhances social presence

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through the sense of embodiment (Kilteni *et al.*, 2012)—comprising self-location, agency, and body ownership—which fundamentally alters how participants perceive themselves and others in virtual spaces. This embodied experience has significant implications for equity of voice in virtual meetings, as research indicates that immersive technologies can create more democratic participation patterns by reducing traditional barriers such as interruption apprehension and status-based communication hierarchies that often characterize conventional video conferencing (Greenwald *et al.*, 2017; Kreamer *et al.*, 2024).

Preliminary research indicates that VR meetings can enhance participants' sense of social presence (Smith and Neff, 2018; Yasuoka *et al.*, 2022) and improve task performance, particularly in collaborative activities such as design reviews (Umair *et al.*, 2022). These findings suggest that VR's ability to create more engaging and presence-rich environments could potentially address the engagement challenges identified in traditional video conferencing platforms.

However, the empirical evidence supporting VR's effectiveness in authentic organizational settings remains limited (see Orel, 2022). Most existing research relies on controlled laboratory experiments with student participants, creating a gap in our understanding of how VR meetings function in real business/professional environments with established teams. Traditional methodologies for assessing meeting effectiveness—whether through self-reported surveys or sensor-based data collection—have proven inadequate for capturing the complex dynamics of VR interactions in professional settings (Frank *et al.*, 2016; Leach *et al.*, 2009). This gap is particularly problematic as organizations continue to invest in VR technologies (see Saghafian *et al.*, 2021; Raji *et al.*, 2024) without robust empirical evidence to guide implementation strategies or justify investments (Kaplan *et al.*, 2021). Given VR's potential to support social presence and engagement, there is a justifiable need for research that examines its effectiveness through the lens of SPT, particularly in real-world organizational contexts where team dynamics and established relationships play epoch-making roles in meeting (favorable) outcomes.

3. Research approach

3.1 Research gap and study rationale

The preceding literature review reveals a specific and critical gap at the intersection of three well-established research streams. While SPT provides a robust framework for understanding engagement in virtual environments (Short *et al.*, 1976; Kreijns *et al.*, 2022), and extensive research documents the engagement challenges of traditional video conferencing (Bailenson, 2021; Standaert *et al.*, 2022), there remains a fundamental disconnect between VR's theoretical potential for enhancing social presence and empirical evidence of its effectiveness in authentic workplace settings. Specifically, existing VR research suffers from a methodological paradox: laboratory studies with student participants suggest VR's superiority over video conferencing (Smith and Neff, 2018; Yasuoka *et al.*, 2022), yet these findings cannot be validated in real organizational contexts due to the inadequacy of current measurement approaches. Traditional engagement measurement methods—whether self-reported surveys or sensor-based approaches—are either too subjective or too intrusive for authentic workplace implementation (Frank *et al.*, 2016; Leach *et al.*, 2009). This creates a critical evidence gap precisely where organizations need it most: understanding whether VR's investment justifies its promised engagement benefits in real professional teams. This gap is particularly acute because SPT suggests that VR should enhance engagement through observable behavioral indicators (Sivunen and Nordback, 2014); yet no study has developed a non-intrusive methodology capable of capturing these behaviors in authentic organizational settings while simultaneously comparing VR to traditional video conferencing.

This identified gap leads to our central research question: How can we empirically measure and explain VR's impact on meeting engagement compared to traditional video conferencing in authentic organizational settings using observable behavioral indicators grounded in SPT?

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3.2 Research design

Our study employs a mixed-methods approach to demonstrate and measure how VR affects meeting engagement compared to traditional video conferencing. Following Schroeder *et al.*'s (2006) recommendation for combining quantitative and qualitative methodologies, we integrate social network analysis with conversation analysis to provide this explanatory power. The social network analysis uncovers the engagement structure within the observed meetings, while the conversational analysis provides insights into content behind those structures contextualizing the structural results (Crossley, 2021). Further building on SPT, the conversation content explains how the structures are built. Providing the causal explanation of the structure through content enables us to better generalize the results of this study, as we can explain why and how the engagement structures differ across modalities.

The study's conceptual framework operationalizes meeting engagement through two primary dimensions: group cohesion and content characteristics. We define meetings as communicative events where participants discuss, negotiate, present, and create material jointly (Frank *et al.*, 2016), with engagement manifested through meaningful communicative actions that advance meeting objectives. This behavioral approach reflects both intrinsic engagement and its materialization (Dobrian *et al.*, 2011), particularly crucial in virtual settings where traditional engagement cues are limited (Said and Carcani, 2020).

3.3 Data collection approach

We conducted a longitudinal experimental study spanning 16 weeks in a large multinational company, comparing four VR-enabled meetings with four video-conferencing meetings. This longitudinal approach, recommended by Gilson *et al.* (2015) and Blanchard and McBride (2020), helped minimize novelty and observation bias.

Our sample comprised an established Scrum team within a multinational technology company comprising six software developers, a team manager, and a Scrum Master. This is a typical agile development team structure, making the experience transferable across agile development teams. As the study's intention is to focus on different modalities of the "new normal," i.e. hybrid work, the intention was to find a hybrid team with limited face-to-face interactions. That is why we selected a team with one member permanently placed in another country, and other members being free to work completely from home. Another condition was to select an established team to avoid fluctuations in engagement and social presence due to progressing through different team formation stages.

We specifically focused on retrospective meetings, which are crucial for team improvement and reflection (Schwaber and Sutherland, 2020) but often face engagement challenges (Przybylek and Kotecka, 2017). This setting was particularly relevant as virtual teams frequently encounter barriers to effective collaboration, including cultural differences, language barriers, and technical difficulties (Gilson *et al.*, 2015). The team's composition and its regular retrospective practices provided an ideal context for examining engagement patterns across different virtual meeting modalities. These sampling characteristics – alongside with other procedures behind the study's data collection – is summarized in the following Figure 1.

Data collection spanned 16 weeks, during which we recorded and transcribed 8 meetings, in the following order: 2 video conferences, 4 VR meetings, and 2 video conferences, using Dovetail 2.0, a qualitative research tool selected for its compliance with corporate data security requirements. Total meeting recordings amounted to 7 h, evenly split between VR (3.5 h) and video conferencing (3.5 h) through Microsoft Teams. VR meetings were conducted using a VR platform (Meta Horizon), with participants using Oculus Quest 2 provided by the company. To ensure comparative validity, we controlled for meeting duration with matched pairs: two 60-min pairs, one 35-min pair, and one 45-min pair. This matching strategy minimized temporal variations while maintaining natural meeting rhythms.

The data collection process followed a systematic protocol to ensure consistency across all sessions. Each meeting was recorded with participant consent and in compliance with

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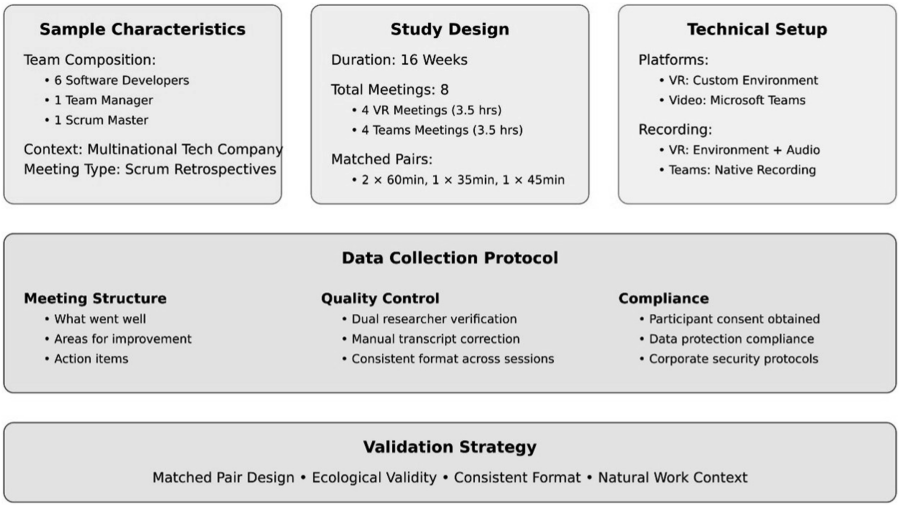

Figure 1. Data collection process. Source: Own

corporate data protection policies. We captured both the virtual environment interactions and audio communications for VR meetings, while video conferences were recorded through Microsoft Teams' native recording feature. The recordings were then transcribed using Dovetail 2.0's automated transcription service, followed by manual verification and correction by two independent researchers to ensure accuracy. All meetings followed the standard Scrum retrospective format, including three main phases: what went well, what could be improved, and action items for the next sprint, allowing for consistent comparison across sessions while maintaining ecological validity in the team's natural work context.

3.4 Data analysis procedures

Our analytical framework incorporates structural and content-based approaches to examine meeting engagement exhaustively. The structural analysis employs social network analysis operating at three distinct levels: group, dyad, and individual. Using SocNetV software, we constructed directed, weighted networks where nodes represent participants and edges capture conversational flows. Network density measures overall group cohesion, reciprocity captures the bidirectional nature of dyadic interactions, and degree prestige evaluates individual engagement levels (Borgatti *et al.*, 2013). This multi-level approach aligns with SPT's emphasis on interactive dynamics while providing quantifiable metrics for comparing engagement patterns across meeting modalities.

We developed a systematic coding scheme grounded in SPT for the content analysis component. Following Andersson *et al.*'s (2007) methodology, we defined a "turn" as a complete communicative act that ends when the speaker yields to another participant, distinguishing it from general speaking instances that include non-meaningful utterances. Our coding scheme identified eight categories of presence-inducing behaviors and expressions, detailed in Table 1, including empathy and emotions, inclusion and interdependence, encouragement, self-disclosure and fun, assistance, attention, positive feedback, and respect and understanding. These categories were derived from established literature on social presence and engagement (Sivunen and Nordback, 2014; Wei *et al.*, 2012; Gorham, 1988).

Complementing the presence-inducing categories, we developed corresponding absence-inducing categories, presented in Table 2, representing behaviors that potentially diminish social presence. These include expressions of feeling alone/refusing interdependence,

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Table 1. Social presence indicators: Behavioral categories and descriptions in virtual meetings

Code	Description	Sources
Empathy and emotions	empathy to others, asking about feelings, acknowledging feelings, showing emotional closeness, providing emotional support, a warm and comfortable relationship with others	Sivunen and Nordback (2014), Wei *et al.* (2012)
Inclusion and interdependence	showing and noticing not being alone, awareness of the other, addressing others by name, referring to us/we, team rituals, showing or asking for interdependence with others, talking about things in common	Sivunen and Nordback (2014), Gorham (1988), Wei *et al.* (2012)
Encouragement	encouraging to talk, asking questions, suggesting follow up	Gorham (1988), Mottet and Richmond (1998), Wei *et al.* (2012), Moallem (2015)
Self-disclosure and fun	personal examples, humor, fun	Gorham (1988), Mottet and Richmond (1998), Moallem (2015)
Assistance	providing and accepting assistance, mutual problem-solving, detailed explanations, and feedback to promote understanding by others	Sivunen and Nordback (2014)
Attention	showing attention to the others, responsiveness	Sivunen and Nordback (2014), Mottet and Richmond (1998), Wei *et al.* (2012)
Positive feedback	praise, compliments, caring, appreciation, positive themes	Gorham (1988), Mottet and Richmond (1998)
Respect and understanding	acknowledging another point of view, insight into the intentions, motivation, and thoughts of the other, respecting other opinions	Sivunen and Nordback (2014), Wei *et al.* (2012), Moallem (2015)

Source(s): Own

misunderstandings and disagreement, task-focus, negative feelings and frustrations, low attention, negative feedback, individual problem-solving, and discouragement. This granular analysis was complemented by a broader examination of discussion themes, recognizing that task-focused discussions typically indicate lower social presence (Mottet and Richmond, 1998), while informal and collaborative problem-solving corresponds to higher social presence indicators.

3.5 Validity and reliability

We implemented a comprehensive validation strategy across multiple dimensions to ensure methodological rigor and data quality. Two independent coders processed all meeting transcripts, establishing initial inter-rater reliability through pilot coding of 20% of the data. This process yielded an inter-rater reliability coefficient of 0.82 (Cohen's κ), with disagreements resolved through structured discussion until consensus was reached. The data cleaning process revealed significant platform differences in non-meaningful speaking instances (33.3% in video conferences versus 3.3% in VR), which were systematically accounted for in subsequent analyses to ensure comparable measurements across platforms.

Our longitudinal design added to validity by allowing participants to acclimate to both meeting modalities over the 16-week study period. Participants demonstrated diminished awareness of being studied by meeting, suggesting natural behavior patterns had emerged. This approach addresses a significant limitation in existing research, which relies on ad hoc student teams in laboratory settings (Anderson *et al.*, 2007). However, we acknowledge potential limitations in generalizability due to our focus on a single team within one organizational context. The matched-pair design for meeting duration and consistent participant composition across modalities strengthens internal validity. However, future

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Table 2. Social absence indicators: Behavioral categories and descriptions in virtual meetings

Code	Description	Sources
Feeling alone, refusing interdependence	showing isolation and seclusion, addressing others impersonally, maintaining distance, refusing to acknowledge mutual dependence, avoiding collaborative interactions	Sivunen and Nordback (2014), Wei *et al.* (2012)
Misunderstandings and disagreement	rejecting others' viewpoints, dismissing contributions, showing lack of insight into others' intentions, expressing confusion, failing to reach shared understanding	Sivunen and Nordback (2014)
Task focus	focusing exclusively on procedures and task completion, avoiding social interaction, maintaining purely transactional communication, bypassing relationship building	Mottet and Richmond (1998)
Negative feelings and frustrations, disrespect	displaying lack of empathy, dismissing emotional expressions, maintaining detached relationships, creating emotional discomfort, showing disrespect, engaging in unfair treatment	Sivunen and Nordback (2014), Mottet and Richmond (1998), Wei *et al.* (2012)
Low attention	demonstrating minimal awareness of others, showing inattentiveness, interrupting speakers, delayed responses, ignoring contributions, providing minimal acknowledgment	Sivunen and Nordback (2014)
Negative feedback	providing harsh criticism, offering unconstructive comments, diminishing others' contributions, expressing demoralizing responses, avoiding improvement suggestions	Gorham (1988), Mottet and Richmond (1998)
Individual problem solving	avoiding assistance offers, working in isolation, refusing help, maintaining independent problem-solving approach	Sivunen and Nordback (2014)
Discouragement	preventing others' participation, providing abrupt responses, blocking discussion continuation, limiting others' speaking opportunities	Mottet and Richmond (1998)

Source(s): Own

research across multiple teams and organizations would be valuable for establishing broader generalizability.

4. Analysis

4.1 Network analysis findings

The data analysis revealed consistently higher engagement levels in VR meetings than traditional video conferencing across all three measured dimensions: group, dyadic, and individual. The following Table 3 summarizes the key network metrics across both platforms.

At the group level, VR meetings demonstrated markedly higher network density across all interaction thresholds. When considering all interactions, VR meetings achieved a network density of 0.795 compared to 0.651 in MS Teams meetings, representing a 22% improvement. This difference becomes more pronounced when examining stronger interaction patterns. For interactions occurring at least twice, VR meetings maintained a network density of 0.597 versus 0.482 in MS Teams (23.65% higher), while for interactions occurring three or more times, the difference expanded further to 31.37% (0.502 in VR versus 0.382 in MS Teams).

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Table 3. Network analysis metrics comparison

Metric level	Measure	VR	MS teams	Difference (%)
Group	Network Density (all ties)	*0.795*	*0.651*	*+22.00%*
	Network Density (2+ ties)	*0.597*	*0.482*	*+23.65%*
	Network Density (3+ ties)	*0.502*	*0.382*	*+31.37%*
Dyadic	Arc Reciprocity	*0.255*	*0.251*	*+1.45%*
	Dyad Reciprocity	*0.278*	*0.247*	*+12.31%*
Individual	Most Prominent Prestige	*0.612*	*0.592*	*+3.38%*
	Least Prominent Prestige	*0.198*	*0.144*	*+37.68%*

Source(s): Own

This progressive increase in density differential with stronger tie requirements suggests that VR environments are particularly effective at fostering sustained, meaningful interactions among participants rather than just superficial exchanges.

At the dyadic level, our analysis reveals more nuanced improvements in interaction patterns. VR meetings demonstrated higher reciprocity measures, with arc reciprocity showing a modest but consistent improvement of 1.45% (0.255 vs. 0.251). More significantly, dyad reciprocity in VR meetings was 12.31% higher (0.278 vs. 0.247), indicating that participant pairs were more likely to engage in two-way communication. This increased dyadic reciprocity suggests that VR environments may be more effective at breaking down communication barriers and encouraging mutual engagement between participants.

Perhaps the most striking findings emerge at the individual level, where degree prestige metrics reveal VR's capacity to enhance participation across different participant types. While the most active participants showed a modest 3.38% increase in degree prestige, the least active participants demonstrated a dramatic 37.68% increase in their degree prestige scores, suggesting that VR environments are particularly effective at encouraging participation from individuals who might otherwise remain peripheral to meeting discussions. These network analysis findings are supported by raw interaction data showing consistently higher interaction frequencies in VR meetings across all measured dimensions, with more balanced participation patterns and fewer instances of extended monologues or dominated discussions. The combination of higher network density, improved reciprocity, and more balanced individual participation metrics suggests that VR creates a fundamentally different interaction environment compared to traditional video conferencing—one that appears to support equity of voice by being more conducive to inclusive and engaged participation by creating a more egalitarian space where traditionally less vocal participants feel more comfortable engaging in discussions.

4.2 Content analysis results

The second part with content analysis uncovered significant distinctions in communication patterns between VR and traditional video conferencing environments, illuminating how immersive technology shapes interaction dynamics and participant engagement. The analysis encompassed both quantitative metrics and qualitative indicators of social presence, revealing systematic differences in how participants communicate and engage across platforms. Table 4 below synthesizes these findings, presenting a comparison of content-related metrics demonstrating distinct interaction and engagement patterns between VR and MS Teams environments.

VR meetings demonstrated a markedly different interaction profile, with 68.42% of instances coded as presence-inducing compared to 64.36% in MS Teams. More striking is the substantial reduction in absence-inducing instances in VR meetings (31.58 versus 56.23% in

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Table 4. Content analysis comparison

Content category	VR meetings	MS teams	Difference (%)
Presence-inducing instances	*68.42%*	*64.36%*	*+6.31*
Absence-inducing instances	*31.58%*	*56.23%*	*−43.80*
Task-oriented topics	*53.30%*	*79.38%*	*−32.85*
Non-task topics	*46.70%*	*20.62%*	*+126.48*
Speaking instances per minute	*5.27*	*3.30*	*+59.70*
Total meaningful instances	*163.45*	*100.00*	*+63.45*
Average turn duration (seconds)	*18.3*	*25.7*	*−28.79*

Source(s): Own

MS Teams), representing a 43.8% decrease in behaviors that potentially diminish social presence. These absence-inducing behaviors included instances of disconnection, misunderstandings, and lack of attention, which occurred significantly less frequently in the VR environment.

On another note, the distribution of discussion topics revealed another significant distinction between the platforms. While MS Teams meetings were dominated by task-oriented topics (79.38% of discussion time), VR meetings maintained a more balanced distribution, with only 53.3% of time devoted to purely task-oriented discussions. Rather than indicating lower efficiency, this reduced task focus in VR appears to create space for relationship-building and informal interactions that contribute to stronger group cohesion. According to SPT, these informal exchanges facilitate progression from basic spatial co-presence to the intersubjective level of social presence, where behavioral engagement emerges through dynamic, responsive interactions that strengthen psychological involvement and create the foundation for sustained collaboration (Biocca *et al.*, 2003; Kreijns *et al.*, 2022). The higher proportion of non-task-related discussions (46.70% in VR compared to 20.62% in MS Teams) coincided with increased presence-inducing behaviors and more dynamic interaction patterns.

The quantitative aspects of interaction patterns further support these qualitative differences, with VR meetings generating 63.45% more meaningful speaking instances overall (5.27 per minute compared to 3.3 in MS Teams) and 28.79% shorter average turn duration (18.3 s versus 25.7 s), suggesting more dynamic and engaged discussions. This pattern explains the difference in absence versus presence coding, where VR turns could be associated with single categories while longer MS Teams turns were more frequently coded as both. These findings point to a fundamental shift in interaction dynamics within VR environments, characterized by more balanced, frequent, and socially present exchanges. The combination of structural and content analysis results suggests that VR technology not only alters how participants interact but also influences what they discuss and how they engage with each other, raising important questions about the mechanisms through which VR enhances engagement and the implications for virtual team collaboration that the following discussion section explores in detail.

5. Discussion

The findings provide evidence that VR technology transforms meeting engagement dynamics compared to traditional video conferencing. Through the methodological framework grounded in SPT, we demonstrate that VR meetings generate significantly higher engagement levels through three key mechanisms: enhanced group cohesion, increased social presence, and more balanced participation patterns. The network density differential between platforms (22–31%) progressively increases with interaction strength, suggesting

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that VR particularly excels at fostering sustained, meaningful exchanges rather than superficial interactions. Furthermore, the dramatic improvement in the least prominent participants' engagement (37.68% increase in degree prestige) indicates VR's potential to democratize virtual meeting participation, addressing a long-standing challenge in remote collaboration.

These findings both support and extend existing theoretical frameworks in several ways. First, our framework contributes significantly to SPT by operationalizing observable behaviors and providing quantifiable metrics for qualitative and quantitative studies. The observed increase in presence-inducing behaviors (68.42% in VR versus 64.36% in MS Teams) aligns with Short *et al.*'s (1976) theoretical foundations, while the reduced prevalence of absence-inducing instances (−43.80%) suggests that VR may actively mitigate the social disconnection typically associated with virtual meetings (Bailenson, 2021). However, our results challenge Driskell *et al.*'s (2003) assertion that virtual environments inherently encourage social loafing. Instead, immersive technology can augment group cohesion and individual accountability, particularly when supported by appropriate social presence affordances.

The study's methodological framework addresses two critical limitations in current collaborative virtual environments research: the reliance on self-reported, post-hoc surveys and the burden of sensor-based studies. Combining social network analysis with conversation analysis provides a more objective, behavior-based approach that can be applied in real-world settings without disrupting natural interaction patterns. This approach moves beyond traditional studies focused on ad hoc student teams in laboratory settings (Anderson *et al.*, 2007), offering a pathway for examining authentic workplace interactions. The framework's multi-level analysis (group, dyad, and individual) provides a more comprehensive understanding of engagement dynamics than existing methods focused solely on individual-level measurements.

Perhaps the most unexpected and thus markable finding was the inverse relationship between task-focus and overall engagement, which provides compelling evidence for SPT's emphasis on the intersubjective level of presence. While MS Teams meetings exhibited higher task-orientation (79.38 versus 53.30% in VR), this came at the cost of reduced social presence and lower overall engagement. According to SPT, informal interactions in VR meetings—often considered "off-task"—facilitate the transition from basic co-presence awareness to deeper psychological involvement and behavioral engagement (Short *et al.*, 1976; Biocca *et al.*, 2003). These informal exchanges create the intersubjective responsiveness that Kreijns *et al.* (2022) identify as essential for sustained social presence, where participants develop behavioral interdependence through dynamic, socially-oriented interactions. This theoretical insight suggests that traditional assumptions about meeting efficiency (Leach *et al.*, 2009) may overlook the fundamental role that informal communication plays in establishing the social presence foundation necessary for effective virtual collaboration.

Our theoretical framework (see Figure 2 below) integrates SPT with empirical findings to explain how VR technology transforms meeting engagement through theoretically grounded mechanisms, providing theoretical backing for the empirical findings presented above. The framework operates across five interconnected levels, beginning with SPT's three-tier structure (Short *et al.*, 1976; Biocca *et al.*, 2003) as the theoretical foundation. VR's technological affordances—immersive 3D environments, avatar-based interaction, spatial audio, and natural gesture recognition—specifically target each level of social presence, from spatial co-presence through psychological involvement to intersubjective behavioral engagement. VR's engagement enhancement occurs through three theoretically derived mechanisms corresponding to social presence manifestations. Enhanced group cohesion emerges through VR's support for intersubjective presence, creating stronger network structures and reciprocal interactions. Strengthened social presence behaviors manifest as increased presence-inducing actions and reduced absence-inducing behaviors, reflecting Sivunen and Nordback's (2014) behavioral indicators. Dynamic interaction balance represents

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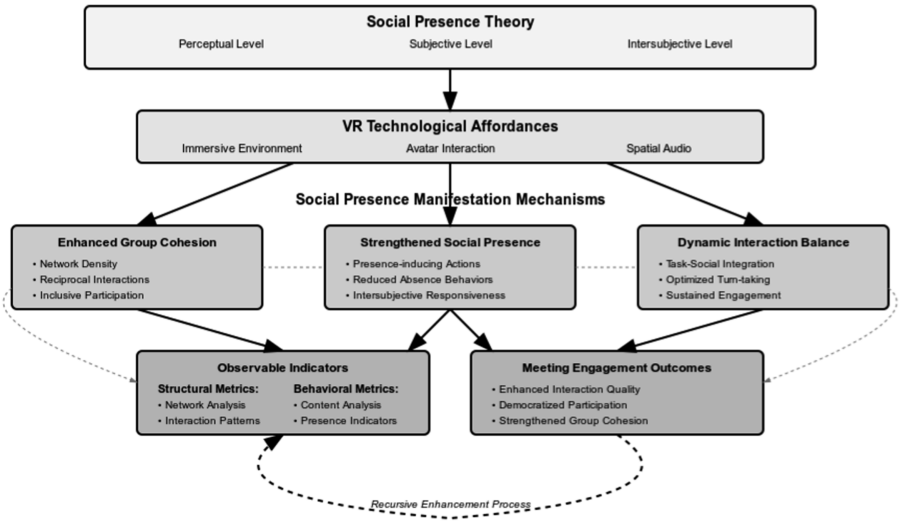

Figure 2. Model of VR-enabled meeting engagement. Source: Own

the task-social integration essential for sustained virtual engagement. Critically, our framework demonstrates how these mechanisms generate both structural effects (measurable through network analysis) and behavioral effects (observable through conversation analysis), providing a methodological bridge between theoretical constructs and empirical measurement. The recursive feedback loops capture social presence's dynamic nature, where behavioral changes reinforce structural transformations, creating a self-sustaining engagement cycle.

The framework's empirical validation demonstrates that VR technology operates as a social presence amplifier rather than merely a more immersive communication medium. The observed increase in network density and improvement in peripheral participant engagement directly correspond to the framework's prediction that VR's technological affordances would enhance intersubjective presence through behavioral engagement mechanisms. The reduction in task-focus in VR meetings, rather than indicating inefficiency, reflects the framework's emphasis on dynamic interaction balance as essential for sustained social presence. Our framework contributes to SPT by operationalizing its abstract constructs through measurable behavioral and structural indicators, addressing the long-standing challenge of empirically validating social presence in authentic organizational contexts. The framework also extends existing virtual collaboration research by demonstrating how technological affordances systematically influence social presence mechanisms, providing organizations with a theoretically grounded roadmap for VR implementation that maximizes engagement outcomes through intentional design of presence-supporting features.

6. Conclusion, limitations, and implications for future research

Through three interrelated mechanisms, our research shows that immersive VR settings thoroughly rearrange the dynamics of meeting involvement: amplified network density patterns that intensify with interaction strength, democratized participation that particularly benefits traditionally peripheral participants, and a recalibrated balance between task-oriented and social interactions that challenges conventional assumptions about virtual meeting efficiency.

Our findings herald a fundamental paradigm shift that transcends traditional physical and virtual collaboration boundaries, establishing a new theoretical foundation for understanding

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human interaction in digitally mediated environments. Rather than viewing VR as an incremental improvement over video conferencing, our SPT-based framework reveals that immersive technologies create entirely new forms of organizational interaction characterized by democratic participation, enhanced social presence, and restructured communication dynamics. This change in thinking demands a reconceptualization of virtual collaboration from technology-centric implementations to human-centric designs that leverage embodied social presence as a strategic organizational capability. While VR technology adoption is still influenced by technological, economic and usability barriers, organizations should recognize that the future of knowledge work lies not in choosing between physical, virtual, or hybrid models, but in understanding how different technological affordances systematically influence social presence mechanisms to create optimal collaboration experiences for specific organizational contexts and objectives.

While our single-team longitudinal study provides rich insights into the transformative potential of VR technology in professional settings, specifically hybrid agile development teams, it also highlights the need for broader investigation across diverse organizational contexts and team configurations. The framework's effectiveness in capturing nuanced interaction patterns suggests promising applications beyond our specific case study, though its generalizability across different organizational cultures and team dynamics requires further validation. The methodological approach we have developed, combining quantitative network analysis with qualitative content examination, offers a robust foundation for future research while acknowledging the complexities inherent in studying virtual collaboration. Our findings regarding the inverse relationship between strict task-focus and overall engagement particularly warrant further investigation across different team types and organizational contexts. What is more, the improvements in peripheral participant engagement suggest intriguing possibilities for using VR to create more inclusive virtual collaboration environments, though the mechanisms behind this effect require deeper examination across diverse team compositions and organizational hierarchies.

The future research agenda arising from our findings opens several intriguing avenues for investigation. How do different organizational cultures moderate the relationship between immersive technologies and meeting engagement? What role does team tenure and familiarity play in the effectiveness of VR-enabled collaboration? How might varying degrees of technological immersion affect the balance between task and social interactions? Future studies should examine these questions while exploring the scalability of VR solutions across larger organizational contexts, the impact of different VR platform designs on engagement patterns, and the potential for VR to address long-standing challenges in cross-cultural virtual collaboration. Particularly promising areas include investigating how VR might transform power dynamics in virtual meetings, the role of avatar personalization in fostering social presence, and the potential for VR to create new forms of collaborative experiences that transcend traditional meeting paradigms. Ultimately, this research positions immersive collaboration not as a technological substitute for face-to-face interaction, but as a fundamentally new medium for human connection that, when properly designed and implemented, can enhance rather than diminish the richness of organizational collaboration. The paradigm shift we document suggests that the future of work will be characterized not by the dominance of any single interaction modality, but by the intelligent orchestration of multiple presence-supporting technologies to create collaboration experiences that surpass what any single medium—physical or virtual—can achieve alone.

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Corresponding author
Veronika Šlapáková Losová can be contacted at: xlosv01@vse.cz

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